

Submatrices of A, B_f and Q_f :-

$$A = [A_t : A_L]$$

$$B_f = [B_{ft} : B_{fL}] ; B_f = [B_{ft} : U]$$

$$Q_f = [Q_{ft} : Q_{fL}] ; Q_f = [U : Q_{fL}]$$

Inter-relation among various matrices :-

① Relation b/w A & B_f :-

For a graph, it can be shown that two matrices A & B_f are orthogonal to each other. i.e

$$AB_f^T = 0$$

$$[A_t : A_L] \cdot \begin{bmatrix} B_{ft}^T \\ \vdots \end{bmatrix} = 0$$

$$A_t B_{ft}^T + A_L = 0$$

$$B_{ft}^T = -A_t^{-1} A_L$$

$$B_{ft} = \left[-A_t^{-1} A_L \right]^T \rightarrow \text{①}$$

② Relation b/w A & Q_f :-

②

For a graph, we know that KCL eqns can be obtained using Incidence as well as f-cutset matrices as

$$A I_b = 0 \rightarrow \textcircled{A}$$

$$\text{and } Q_f I_b = 0 \rightarrow \textcircled{B}$$

where $I_b = \begin{bmatrix} I_t \\ \vdots \\ I_l \end{bmatrix}$, branch current matrix of $b \times 1$

$$\text{So eqn } \textcircled{A} \quad [A_t : A_l] \begin{bmatrix} I_t \\ \vdots \\ I_l \end{bmatrix} = 0$$

$$A_t I_t + A_l I_l = 0$$

$$I_t = -A_t^{-1} A_l I_l \rightarrow \textcircled{C}$$

eqn $\textcircled{B}$ becomes,

$$[U : Q_{fl}] \begin{bmatrix} I_t \\ \vdots \\ I_l \end{bmatrix} = 0$$

$$I_t + Q_{fl} I_l = 0$$

$$I_t = -Q_{fl} I_l \rightarrow \textcircled{D}$$

From eqn $\textcircled{C}$ & $\textcircled{D}$,

$$\boxed{Q_{fl} = A_t^{-1} A_l} \rightarrow \textcircled{E}$$

③ Relation b/w B_f & Q_f :- The reln can be found out by regns $\textcircled{D}$ & $\textcircled{E}$

$$\boxed{Q_{fl} = -B_{ft}^T \text{ or } B_{ft} = -Q_{fl}^T}$$

Relationship Among Parameters :-

③

For a graph,

$$\text{KVL :- } B_f V_b = 0 \rightarrow \textcircled{A}$$

$$\text{KCL, } A I_b = 0 \rightarrow \textcircled{B}$$

$$\text{and } Q_f I_b = 0 \rightarrow \textcircled{C}$$

$$\text{where } V_b = \begin{bmatrix} V_t \\ \vdots \\ V_l \end{bmatrix}, I_b = \begin{bmatrix} I_t \\ \vdots \\ I_l \end{bmatrix}$$

Relation b/w link voltages and twig voltages :-

From Eqn (A),

$$[B_{ft} : U] \begin{bmatrix} V_t \\ \vdots \\ V_l \end{bmatrix} = 0$$

$$B_{ft} V_t + V_l = 0$$

$$V_l = -B_{ft} V_t \rightarrow \textcircled{1}$$

$$V_l = Q_{fl}^T V_t \rightarrow \textcircled{2}$$

Relation between Branch Voltages & Twig Voltages :-

From eqn (2),

$$V_b = \begin{bmatrix} V_t \\ \vdots \\ V_l \end{bmatrix} = \begin{bmatrix} U V_t \\ \vdots \\ Q_{fl}^T V_t \end{bmatrix}$$

$$V_b = Q_f^T V_t \rightarrow \textcircled{3}$$

Relation b/w Branch voltages & Node Voltages

$$V_b = A^T V_n \rightarrow \textcircled{4}$$

↳ voltage of various nodes.

Relation b/w twig currents & link currents :- (4)

From Relation b/w A & Q_f and Relation b/w Q_f & Q_l

$$I_t = -A_t^{-1} A_l I_l$$

$$= -Q_{fl} I_l = B_{ft}^T I_l \rightarrow (5)$$

Relation b/w Branch currents & link currents :-

$$I_b = \begin{bmatrix} I_t \\ \vdots \\ I_l \end{bmatrix} = \begin{bmatrix} B_{ft}^T I_l \\ \vdots \\ I_l \end{bmatrix} = \begin{bmatrix} B_{ft}^T \\ \vdots \\ I \end{bmatrix} I_l$$

$$I_b = B_f^T I_l \rightarrow (6)$$

NETWORK ANALYSIS:-

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for a general n/w with many branches, branch voltages & branch currents can be written as

$$V_b = V_s + Z_b(I_b - I_s) \rightarrow (1)$$

$$I_b = I_s + Y_b(V_b - V_s) \rightarrow (2)$$

V_b = branch voltage matrix of $b \times 1$

I_b = " current matrix of $b \times 1$

V_s = voltage source matrix of $b \times 1$

I_s = current " " " "

Z_b = branch impedance matrix of $b \times b$

Y_b = branch admittance matrix of $b \times b$

Loop Analysis:-

The branch voltages are given by eqn (1)

$$V_b = V_s + Z_b(I_b - I_s)$$

Premultiplication by B_f ,

$$B_f V_b = B_f V_s + B_f Z_b(I_b - I_s)$$

$$0 = B_f V_s + B_f Z_b(B_f^T I_l - I_s)$$

$$[\text{Since } B_f V_b = 0 \text{ \& } I_b = (B_f^T I_l)]$$

$$B_f Z_b I_s - B_f V_s = B_f Z_b B_f^T I_l$$

$$\text{Let } B_f Z_b I_s - B_f V_s = V_l$$

$$\text{and } B_f Z_b B_f^T = Z_l$$

$$\text{Then } V_l = Z_l I_l$$

This is called matrix loop eqn.

Z_l = loop Impedance matrix, I_l = loop/link current matrix

$$I_l = V_l Z_l^{-1}$$

Nodal Analysis:-

The branch ~~volt~~ currents are given by eqn(2)

$$I_b = I_s + Y_b (V_b - V_s)$$

Premultiplication by A,

$$A I_b = A I_s + A Y_b (V_b - V_s)$$

$$0 = A I_s + A Y_b (A^T V_n - V_s)$$

(Since $A I_b = 0$ and $V_b = A^T V_n$)

$$\underbrace{A Y_b V_s}_{I_n} - A I_s = \underbrace{A Y_b A^T}_{Y_n} V_n$$

$$I_n = Y_n V_n$$

This is called matrix node equation

Y_n = node admittance matrix

V_n = node voltage matrix